

LAVANYA B

Sr. Cloud/DevOps Engineer

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PROFESSIONAL SUMMARY:

- **DevOps Engineer** with 7+ years of experience in **AWS, Azure, GCP, Terraform, Ansible, Docker, Kubernetes, Jenkins, and GitLab CI/CD**, specializing in **automation, cloud migration, CI/CD pipelines, Agile delivery, monitoring, and DevSecOps** to deliver secure, scalable, and resilient enterprise infrastructures.
- Designed and deployed **AWS** infrastructures using **CloudFormation, Auto Scaling Groups, and VPCs**, enabling secure, scalable, and resilient enterprise workloads across **healthcare, fintech, and e-commerce** domains.
- Optimized **AWS** resources including **Load Balancers, EC2, and storage services** to improve application performance and ensure high availability in production environments.
- Migrated and managed enterprise workloads on **Azure** using **ARM templates, VNets, and governance policies**, enforcing compliance with regulatory standards and streamlining provisioning activities.
- Configured **Azure Application Insights, Load Balancers, and scaling rules**, ensuring seamless service performance and reliable transaction handling for business-critical systems.
- Deployed **Azure DevOps** pipelines to automate builds, testing, and deployments, integrating with **Git, Docker, and Terraform** to accelerate delivery and maintain traceable, secure deployments across environments.
- Architected **GCP** disaster recovery and multi-cloud integration solutions, delivering high-availability architecture and uninterrupted business continuity across regulated domains.
- Automated provisioning, scaling, and resource policies in **GCP** using reusable **templates**, ensuring consistency and audit-ready infrastructure across development, staging, and production environments.
- Developed standardized **Terraform** modules for **AWS, Azure, and GCP**, eliminating infrastructure drift, improving deployment speed, and ensuring compliance across enterprise rollouts.
- Automated configuration management and security enforcement with **Ansible playbooks**, streamlining patching, updates, and reducing manual intervention across large-scale **Linux** server environments.
- Containerized applications using **Docker**, building optimized images and integrating them with **CI/CD pipelines** to improve portability, consistency, and accelerated release cycles.
- Orchestrated containerized workloads with **Kubernetes** and **Mesos**, applying health checks, rolling updates, and autoscaling strategies to maintain zero-downtime deployments.
- Modernized legacy systems by migrating workloads to **OpenShift**, enabling secure **container orchestration**, stronger governance, and improved reliability of enterprise applications.
- Built and maintained **Jenkins** pipelines for end-to-end **CI/CD**, integrating **Git, Maven, Docker, Terraform, and SonarQube** to deliver automated, reliable, and repeatable deployment workflows.
- Designed and optimized **GitLab CI/CD** pipelines to support enterprise collaboration, improve governance, and accelerate release cycles across distributed development teams.
- Implemented observability solutions using **Prometheus, Grafana, ELK, and Datadog**, providing actionable insights, ensuring **SLA compliance**, and improving incident resolution.
- Authored **Python** and **Bash** automation scripts for **infrastructure provisioning, log parsing, compliance checks, and server maintenance**, improving efficiency and operational consistency.
- Provisioned and standardized **Linux servers** with **Kickstart**, ensuring consistent builds, timely patch compliance, and secure environments across hybrid cloud infrastructures.
- Embedded **DevSecOps** practices into **CI/CD pipelines** by integrating **SonarQube, Trivy, and OWASP** checks, enforcing secure coding standards and ensuring compliance across environments.

- Designed and implemented **MLOps** pipelines for **fraud detection** and **healthcare analytics**, managing **versioning, deployment, and monitoring** to ensure reproducibility and scalability.
- Leveraged **Generative AI** tools such as **GitHub Copilot** and AI-driven **log summarization** to accelerate script generation, automate documentation, and enhance root-cause analysis.

TECHNICAL SKILLS:

Cloud Platforms	AWS Microsoft Azure Google Cloud Platform (GCP)
Infrastructure as Code (IaC)	Terraform CloudFormation ARM Templates
Configuration Management	Ansible Chef Puppet
CI/CD Tools	Jenkins GitLab CI/CD Azure DevOps Maven CI/CD Pipelines
Containerization & Orchestration	Docker Kubernetes (EKS, AKS, GKE, Mesos) OpenShift
Monitoring & Logging	Prometheus Grafana ELK/EFK Stack Datadog CloudWatch Azure Monitor Application Insights Nagios
Security & Compliance	DevSecOps SonarQube Trivy OWASP SSL/TLS IAM RBAC Vault KMS CloudTrail Compliance (PCI DSS, HIPAA, FDA)
Scripting & Automation	Python Bash Shell PowerShell
Databases	MySQL PostgreSQL Oracle SQL Server MongoDB Cloud DSatabases (AWS RDS, Azure SQL, GCP Cloud SQL)
Operating Systems & Servers	Linux (RHEL, Ubuntu, CentOS, Kickstart) Windows Server Apache Nginx Tomcat
Build & Release Management	Maven Artifactory
MLOps & Generative AI	Kubeflow, MLflow, containerized ML workloads, pipelines, GitHub Copilot, AI log analysis, Generative AI automation

WORK EXPERIENCE

Client: Medtronic | Minnesota, USA

Feb 2025 – Present

Role: Sr. Cloud DevOps Engineer

- Built and supported **AWS infrastructure** including **EC2, S3, IAM, VPC, CloudWatch, and Auto Scaling Groups**, delivering scalable solutions for medical platforms while improving productivity and compliance.
- Deployed and maintained **Azure Kubernetes Service (AKS)** clusters for healthcare workloads, ensuring high availability, reliability, and adherence to regulatory standards.
- Migrated legacy on-premises systems into the **Azure cloud**, enhancing scalability, reducing operational costs, and maintaining consistent compliance across enterprise healthcare platforms.
- Modernized healthcare workloads by migrating them to **OpenShift clusters**, enabling secure container orchestration and improving performance and reliability for production applications.
- Authored reusable **Terraform modules** for AWS and Azure to automate provisioning, which ensured reproducibility, compliance, and consistency across healthcare environments.
- Provisioned secure **VPCs, IAM roles, cloud storage, and databases** with Terraform, reducing configuration drift, preventing misconfigurations, and accelerating environment setup.
- Integrated **Terraform workflows** into **Jenkins** and **GitLab CI/CD pipelines**, streamlining infrastructure rollouts and ensuring environments remained fully audit-ready.

- Developed **Ansible playbooks** for configuration management, patching, and compliance updates across Linux server fleets, reducing manual effort and strengthening system security.
- Automated deployments of clinical data systems using **Ansible roles**, ensuring consistency across environments and meeting **FDA audit** and validation requirements.
- Containerized critical healthcare applications with **Docker**, enabling portability, consistency, and simplified deployments across development, testing, and production.
- Integrated **Docker image builds** into **Jenkins pipelines**, improving release cycles, reducing risks, and strengthening compliance in sensitive healthcare systems.
- Managed **Kubernetes clusters (EKS/AKS)** for regulated workloads, implementing auto-scaling, **RBAC policies**, and network isolation for secure and compliant operations.
- Configured readiness probes, health checks, monitoring, and logging in **Kubernetes**, minimizing downtime and maintaining availability for patient-centric applications.
- Designed and maintained **Jenkins pipelines** to automate builds, testing, and deployments for medical software, embedding compliance validation into all workflows.
- Integrated **Jenkins** with **SonarQube, Docker, and Terraform**, ensuring secure, repeatable, and audit-ready software releases across healthcare projects.
- Built and supported **GitLab CI/CD pipelines**, streamlining collaboration across distributed teams and ensuring traceability in release processes.
- Implemented structured **Git branching strategies** and version control practices, ensuring collaboration, traceability, and proper documentation across DevOps teams.
- Configured **Maven builds** for **Java applications**, producing validated, reproducible results aligned with strict regulatory requirements.
- Supported **Java applications** within DevOps pipelines, ensuring stable builds, smooth deployments, and consistent performance in validated environments.
- Deployed **Prometheus, Grafana, and Datadog dashboards** to monitor application performance and device telemetry, providing actionable insights to engineering and compliance teams.
- Implemented **ELK logging pipelines** with automated alerts, improving incident detection, reducing resolution time, and generating reliable audit reports.
- Wrote **Python** and **Shell scripts** to automate compliance checks, log parsing, system monitoring, and server maintenance, improving accuracy and saving manual effort.
- Leveraged **GenAI tools (GitHub Copilot)** to generate automation scripts and documentation, reducing repetitive work and enhancing productivity.
- Applied **AI-based log analysis** solutions to summarize errors and accelerate root cause analysis, improving incident resolution workflows.
- Partnered with data science teams to containerize and deploy **ML models on Kubernetes**, enabling scalable and reproducible predictive healthcare applications.
- Automated **ML model versioning, monitoring, and CI/CD integration**, ensuring reliable and consistent model deployment pipelines.
- Configured **IAM policies, KMS encryption, CloudTrail logging, and network security controls** to protect sensitive patient and device data in regulated environments.
- Integrated **SonarQube, Trivy, and OWASP scans** into pipelines to enforce secure coding practices and ensure vulnerabilities were addressed early in the release cycle.
- Applied **IAM rules, encryption standards, and container security controls**, aligning DevOps workflows with **HIPAA** and **FDA compliance** requirements.

ENVIRONMENT

AWS | GCP | Terraform | CloudFormation | Jenkins | GitLab CI | Docker | Kubernetes (EKS/GKE) | OpenShift | Mesos | Ansible | Helm | Python | Java | Bash | ELK/EFK Stack | Prometheus | Grafana | Datadog | SonarQube | Trivy | Maven | Artifactory | GenAI | MLOps | DevSecOps

Role: SRE & DevOps Engineer

- Migrated high-volume payment **microservices** to **Azure**, enabling secure global transaction processing while strengthening reliability and disaster recovery for mission-critical services.
- Configured **Azure VNets**, **Load Balancers**, and **Storage Accounts**, optimizing network design to reduce latency and ensuring seamless uptime for core payment pipelines.
- Automated deployments with **ARM templates** and governance policies, ensuring consistent rollouts across **PCI DSS-compliant environments** and reducing risks tied to manual provisioning.
- Monitored production workloads with **Azure Application Insights**, proactively identifying bottlenecks and improving the stability and performance of payment operations.
- Built scalable **CI/CD pipelines in Azure DevOps**, enhancing collaboration between distributed teams and improving release reliability with stronger traceability and governance.
- Developed reusable **Terraform modules** to provision PCI-compliant infrastructure, standardizing deployments across environments and supporting compliance audits.
- Managed multi-region infrastructure rollouts with **Terraform**, accelerating onboarding of new high-volume payment workloads while maintaining infrastructure consistency.
- Automated configuration, patching, and compliance checks with **Ansible playbooks**, ensuring secure, policy-driven environments for sensitive financial systems.
- Containerized legacy payment applications with **Docker**, improving portability, reducing compatibility issues, and streamlining deployments across hybrid cloud environments.
- Managed private **Docker registries** for microservices, optimizing image storage and enabling faster deployments for distributed engineering teams.
- Deployed containerized services on **Kubernetes clusters**, applying autoscaling and rolling updates to maintain zero-downtime even during peak global transaction periods.
- Tuned **Kubernetes/Mesos** resource allocation and scaling policies, optimizing performance under heavy workloads while improving overall cost efficiency.
- Migrated mission-critical payment services to **OpenShift**, enforcing policy-driven deployments and improving governance in highly regulated financial environments.
- Designed automated build and deployment pipelines in **Jenkins**, integrating **Docker** and **Maven** to streamline releases and improve service reliability.
- Integrated **Jenkins** with monitoring and container platforms, enhancing observability, enabling early detection of issues, and supporting faster incident recovery.
- Built **GitLab CI/CD pipelines** for payment and fraud detection workflows, automating testing and deployments to improve reproducibility and secure delivery.
- Maintained **Git repositories** with structured branching strategies and pull requests, fostering collaboration and ensuring codebase integrity across global teams.
- Standardized builds of **Java** payment services with **Maven**, managing dependencies effectively and ensuring artifact consistency across environments.
- Implemented monitoring dashboards with **Prometheus**, **Grafana**, and **Azure Monitor**, providing real-time visibility and supporting continuous uptime.
- Developed custom **PowerShell scripts** to automate administrative and monitoring tasks for **Windows servers**, improving operational efficiency and maintaining compliance.
- Enhanced **.Net microservices** within fraud detection workflows, strengthening security controls and ensuring smooth integration with payment pipelines.
- Tuned **SQL queries** and managed large-scale **databases**, optimizing response times and improving the reliability of payment and fraud detection systems.
- Administered production **Windows servers**, ensuring patch compliance, secure operations, and uninterrupted service availability.
- Implemented automated **MLOps pipelines** for fraud detection models, streamlining training, versioning, and deployment to deliver consistent results.
- Designed **Generative AI** solutions to automate reporting workflows and customer communications, reducing manual effort and increasing operational efficiency.

- Embedded **DevSecOps** practices into **CI/CD pipelines**, integrating vulnerability scans, secrets management with **Vault**, and **KMS-based encryption** to ensure PCI DSS-aligned secure service releases.

ENVIRONMENT:

Azure | Azure DevOps | Terraform | ARM Templates | Ansible | Jenkins | GitLab CI | Git | Docker | Kubernetes (AKS, Mesos) | OpenShift | Maven | .Net | PowerShell | Windows Server | SQL Database | Prometheus | Grafana | Azure Monitor | CI/CD Pipelines | Auto-scaling | Load Balancers | Monitoring & Logging | DevSecOps | MLOps | Generative AI

Client: First American Bank | New Jersey, USA

Jan 2021 – July 2023

Role: Cloud & DevOps Engineer

- Designed and implemented **AWS infrastructure** using **CloudFormation templates**, provisioning secure and scalable banking applications across multiple environments while improving reliability.
- Built and managed **CloudFormation stacks** to deploy applications with standardized configurations, ensuring consistency across regions and minimizing manual intervention.
- Updated **CloudFormation templates** to support scaling, patching, and configuration changes, enabling automated infrastructure updates that strengthened resiliency in banking systems.
- Configured **GCP environments** for disaster recovery, multi-cloud integration, and automated provisioning, ensuring uninterrupted business continuity for critical workloads.
- Optimized **AWS resources** by configuring auto-scaling, load balancers, and storage solutions, balancing operational efficiency with high performance for customer-facing applications.
- Developed and maintained **Terraform scripts** to automate infrastructure provisioning across **AWS and GCP**, enabling compliance, auditable, and repeatable deployments across multiple projects.
- Created reusable **Terraform modules** for standard components, reducing duplication of code and accelerating environment provisioning for regulated banking systems.
- Configured **Jenkins pipelines** to automate build, deployment, and CI/CD workflows, integrating **Git, GitLab, and Maven** for seamless delivery of enterprise applications.
- Enhanced **Jenkins pipelines** with better error handling and optimized build times, enabling faster release cycles while maintaining stability in production.
- Implemented **Chef configurations** to automate provisioning, patch management, and deployments, ensuring secure and compliant server environments across all stages.
- Migrated enterprise workloads to **OpenShift**, containerizing applications to improve portability, standardizing deployments, and strengthened governance in hybrid infrastructures.
- Managed **GitLab repositories** to support secure version control, code reviews, and team collaboration, maintaining traceability and compliance across distributed teams.
- Configured **Git branching strategies** and review processes to streamline development workflows, minimize merge conflicts, and maintain production-ready codebases.
- Integrated **Maven** for automated builds, dependency management, and artifact delivery, ensuring reliable and consistent release processes for **Java applications**.
- Implemented monitoring with **Prometheus, Grafana, and CloudWatch** to track health, performance, and usage metrics, improving observability across all environments.
- Developed **Python scripts** for log parsing, compliance reporting, and infrastructure monitoring, automating repetitive tasks and reducing manual engineering workload.
- Created **Bash scripts** to manage server backups, maintenance scheduling, and routine tasks, improving efficiency and reliability of banking environments.
- Provisioned and maintained **Linux servers** using **Kickstart**, standardizing builds, enforcing timely patching, and ensuring secure operations for production systems.

ENVIRONMENT:

AWS | GCP | Terraform | CloudFormation | Jenkins | GitLab | Git | Maven | Python | Bash/Shell | Linux/Kickstart | Chef | OpenShift | Monitoring Tools

Client: Zensar Technologies | Bangalore, India

Sept 2018 – Dec 2020

Role: DevOps Engineer

- Agile methodologies such as Scrum and Kanban **were applied** to plan, prioritize, and deliver cloud infrastructure features, enhancing collaboration between teams and accelerating release cycles.
- Designed and deployed **AWS infrastructure** with **CloudFormation templates**, building secure, scalable, and highly available **e-commerce platforms** that supported consistent multi-environment rollouts.
- Managed and maintained **CloudFormation stacks** across multiple regions, ensuring infrastructure consistency, minimizing downtime, and supporting reliable performance for production workloads.
- Built and optimized **Docker images**, standardizing environments and accelerating development, testing, and deployment workflows for enterprise-scale applications.
- Configured **Kubernetes clusters** to orchestrate containerized applications, implementing auto-scaling and fault-tolerance strategies to ensure seamless customer-facing services.
- Developed **CI/CD pipelines in Jenkins**, integrating with **Git** for version control and **Maven** for builds, which streamlined deployments and improved overall release reliability.
- Implemented monitoring solutions with **CloudWatch, Prometheus, and Grafana**, providing visibility into application performance, resource utilization, and SLA compliance for mission-critical services.
- Automated infrastructure tasks and **log monitoring** with **Python scripts**, reducing manual intervention and enabling proactive detection of issues in production environments.
- Created **Bash and Shell scripts** for server management, scheduled backups, and maintenance automation, improving reliability and ensuring stability in production systems.
- Provisioned and configured **Linux servers** using **Kickstart**, ensuring standardized builds, timely patching, and secure operations across hybrid environments.
- Documented automation workflows, scripting practices, and operational processes in a centralized repository, enabling smoother onboarding and maintaining operational continuity across teams.

ENVIRONMENT

AWS | CloudFormation | Docker | Kubernetes | Jenkins | Git | Maven | Python | Bash/Shell | Linux/Kickstart | Monitoring Tools | Agile

Client: Sterlite Technologies | Pune, India

Jan 2018 – Aug 2018

Role: Infrastructure Automation Specialist

- Practiced Agile methodologies such as **Scrum** and **Kanban** to plan and prioritize infrastructure tasks, streamlining delivery of **Linux server provisioning**, automation, and **CI/CD** activities for telecom projects.
- Provisioned and standardized 50+ **Linux servers** using **Kickstart**, enabling rapid, consistent deployments across multiple data centers and eliminating repetitive manual configuration steps.
- Installed, configured, and maintained **Linux servers** for large-scale telecom applications, handling patching, kernel tuning, and performance optimization to ensure smooth production operations.
- Created **Puppet** manifests and reusable modules to automate configuration management, software installation, and application setup, reducing errors and ensuring consistency across environments.
- Enhanced and maintained existing **Puppet** scripts, enforcing configuration compliance in production, staging, and testing environments while minimizing drift and deployment issues.
- Designed and managed **Jenkins pipelines** to automate build, deployment, and application testing workflows, accelerating release cycles and improving overall delivery confidence for engineering teams.

- Authored **Bash scripts** to automate system tasks like log monitoring, server maintenance, and backup scheduling, improving operational reliability and minimizing downtime.
- Configured and managed **Nagios monitoring** for **Linux servers** and telecom applications, setting up alerts and dashboards to proactively detect issues and ensure uninterrupted operations.
- Monitored **Linux infrastructure** using system logs, alerts, and reporting tools, proactively identifying issues and supporting uptime for mission-critical telecom platforms.
- Implemented system hardening techniques, applied security patches, and enforced compliance policies across **Linux environments** to strengthen security posture and meet telecom IT standards.
- Documented **Linux administration** and **Kickstart provisioning** procedures in a centralized repository, improving knowledge sharing, onboarding, and continuity for junior engineers and operations teams.

ENVIRONMENT

Puppet | Jenkins | Bash | Linux/Kickstart | Git | Nagios | SSH | SCP | Rsync | Fiber-optic Network Infrastructure

EDUCATION

Concordia University Saint Paul

M.S in Information technology and management

08/2023 - 12/2024